

DecodeLabs Internship

Project 3: Context-Anchored Answering (RAG Basics)

Objective

Prevent AI hallucination by injecting a specific reference document and forcing the model to behave as a Closed-Book Question Answering System. The model must answer only from the provided reference text, include citations, and return 'Information Not Found' when the answer does not exist in the document.

1. Context-Anchored (RAG) Master Prompt

You are a Closed-Book Question Answering System.

Rules:

1. Use ONLY the provided reference text.
2. Do NOT use external knowledge.
3. Do NOT guess.
4. If the answer is not present in the reference text, reply exactly:
Information Not Found
5. Every answer must include the paragraph citation.

Reference Text:

Paragraph 1:
DecodeLabs was founded in 2020.

Paragraph 2:
The company provides AI and Data Science training.

Paragraph 3:
The headquarters is located in Lucknow.

Question:
Where is DecodeLabs headquartered?

Output:

Answer:
Lucknow

Citation:
Paragraph 3

2. Negative Constraint Example

Question:
Who is the CEO of DecodeLabs?

Instruction:
Answer only from the provided reference text.
If information is missing, return:
Information Not Found

Output:

Information Not Found

3. Test Cases

Test Case 1

Input:
Reference:
Paragraph 1: DecodeLabs was founded in 2020.
Paragraph 2: Headquarters is located in Lucknow.

Question:
When was DecodeLabs founded?
Output:
Answer:
2020

Citation:
Paragraph 1

Test Case 2

Input:
Reference:
Paragraph 1: DecodeLabs was founded in 2020.
Paragraph 2: Headquarters is located in Lucknow.

Question:
Where is DecodeLabs headquartered?
Output:
Answer:
Lucknow

Citation:
Paragraph 2

Test Case 3

Input:
Reference:
Paragraph 1: DecodeLabs was founded in 2020.
Paragraph 2: Headquarters is located in Lucknow.

Question:
Who is the CEO of DecodeLabs?
Output:
Information Not Found

Test Case 4

Input:
Reference:
Paragraph 1: Python is a programming language.
Paragraph 2: It was created by Guido van Rossum.

Question:
Who created Python?
Output:
Answer:
Guido van Rossum

Citation:
Paragraph 2

Test Case 5

Input:
Reference:
Paragraph 1: Python is a programming language.
Paragraph 2: It was created by Guido van Rossum.

Question:
When was Python first released?
Output:
Information Not Found

Conclusion

This project demonstrated Context-Anchored Answering using RAG principles. The model was restricted to the supplied reference document, required to provide citations for every answer, and prevented from hallucinating by using strict negative constraints such as returning 'Information Not Found' whenever the requested information was unavailable in the provided text.